

Cobalt-60 Lab Report

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Goal:

Determine if the probability distribution for decays of Cobalt-60 matches a Poisson Distribution.

Procedure:

1. Measure the number of decays in 10-second intervals for 125 trials.
2. Determine the average number of decays in the 10-second intervals. We verified the accuracy of our mean value by taking a recursive average of the number of decays per 10 seconds. Then we graphed the average number at “n” trials versus each trial. The graph levels out to a specific value which we will call the average.
3. Create a histogram of the decays per 10 seconds.
4. Convert the histogram to probability so we can overlay the Poisson distribution on top of the graph.
5. Superimpose a Poisson distribution to the histogram using $\bar{N}$ as the recursive average and treating N as the independent variable in the Poisson Distribution formula:

$$P(N, \bar{N}) = \frac{e^{-\bar{N}} \bar{N}^N}{N!} \quad [1]$$

6. Multiply the Poisson distribution by a scalar so that the maximum of the histogram and the maximum of the Poisson match.
7. Determine the standard error and standard deviation of the Poisson distribution.
8. Verify if the Poisson distribution does in fact match the histogram.

Results:

After recording the decays per 10 seconds for 125 trials, we found that the average decays per 10 seconds was 80.3 ± 0.8 decays with a standard deviation of 8.96 decays. We verified this by checking our recursive average as seen in **Figure 1**. The graph of the recursive average did level out to 80.3 decays per 10 seconds. We used this average in the function of the Poisson distribution [1]. Next we created a histogram of the decays per 10 seconds and converted it to a probability histogram by dividing the data by how many trials we had, in this case 125. With this histogram, we then fit our equation for the Poisson distribution to the probability histogram. We used 4.382 as our scalar multiple to match the maxes of the histogram and the Poisson distribution. We found this by dividing the maximum value of the histogram by the original maximum value of the Poisson distribution to find our scalar value. This can all be seen in **Figure 2**.

Conclusion:

Once we found that our average value for the number of decays per 10 seconds was accurate via the recursive graph, we concluded that our average is 80.3 ± 0.8 decays. For the superimposed graph of our probability histogram and the scaled Poisson distribution, we believe that the Poisson distribution fits our data well, and perhaps with a different bin size it would look a bit cleaner, but the overall fit is still close to our data.

Figure 1: Recursive Average of Decays (per 10s) for 125 Trials

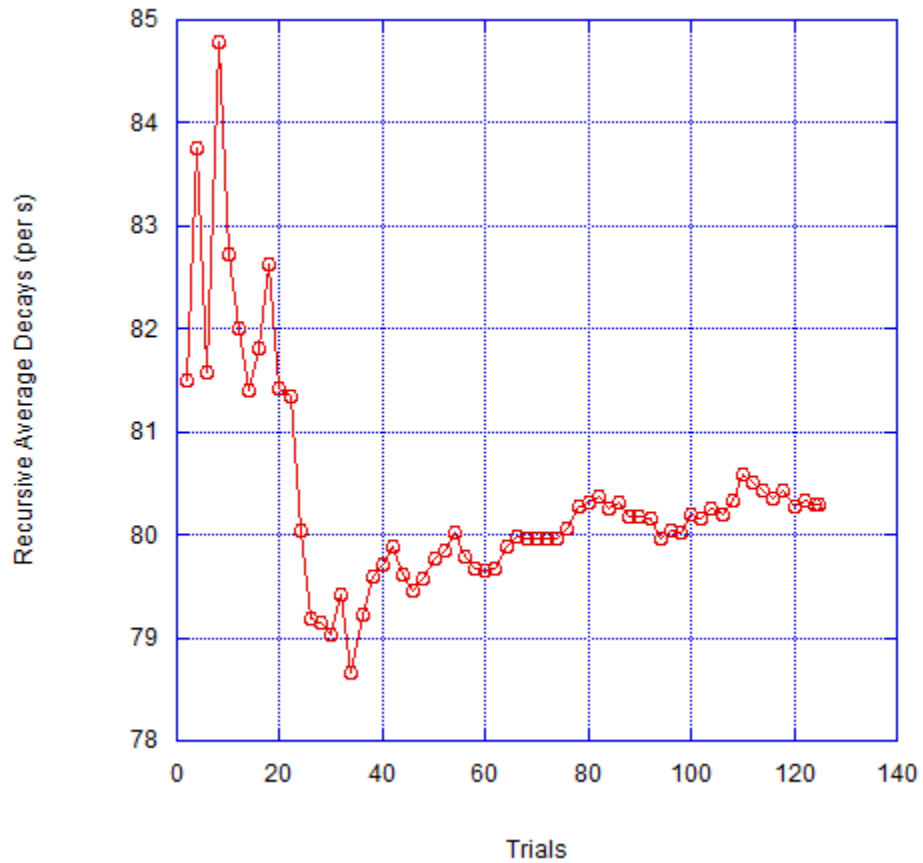


Figure 1 shows that as we add more trials, our average for the number of decays per 10 seconds* for our Cobalt sample evens out around 80.3, and since the deviations from that point always trend back to around 80.3, we can be confident that our average is accurate.

*The y-axis is supposed to be (per 10s) not (per s)

Figure 2: Histogram for Probability of # decays per second, with Poisson distribution (scaled)

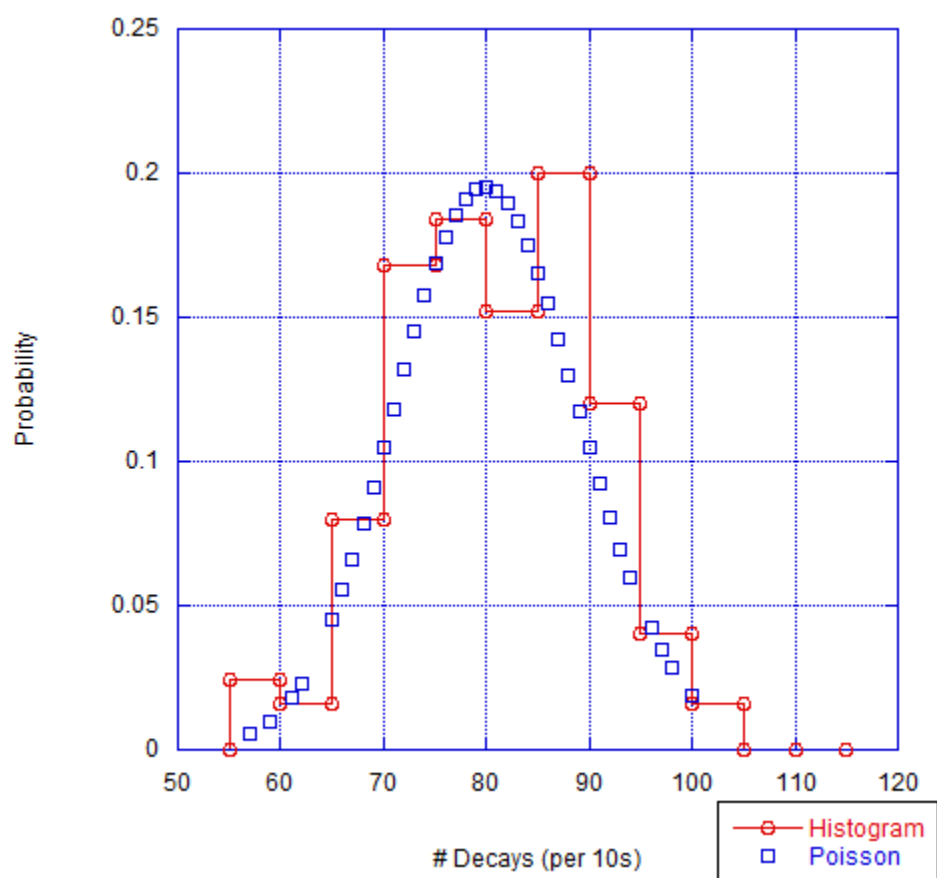


Figure 2 has our histogram converted to probability overlaid with our scaled Poisson distribution.